Waseem Raza, PhD(Electrical and Computer Engineering)

Researcher & Engineering Educator | Intelligent Wireless Systems

Wireless AI | 5G/6G | Network Optimization | Research & Education

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LinkedIn | GitHub | Google Scholar | Coursera Courses | Portfolio

Academic Profile

Electrical and Computer Engineering researcher and educator specializing in AI/ML for wireless networks, resilient learning, and 5G/6G network intelligence. Research spans propagation modeling, generative data augmentation, digital twins, optimization, self-healing, domain adaptation, and reinforcement learning. Academic work includes teaching, laboratory instruction, student mentoring, research publications, proposal development, and curriculum activities. Industrial R&D at AT&T Labs, Futurewei, Impac, and AT&T brings operational data and applied validation to this research program.

Research Interests

AI/ML for 5G/6G Wireless Networks | Robust and Data-Efficient Learning | Network Optimization and Self-Healing | Propagation Modeling and Digital Twins | Transfer Learning and Generative Augmentation | Reinforcement Learning for Autonomous Networks | Wearable Intelligence and Human-Centered Sensing

Doctoral Research Focus

Doctoral research in AI/ML for wireless networks, emphasizing resilient propagation models, data-efficient generative learning, adaptive network optimization, self-healing, and reproducible evaluation.

Education

PhD, Electrical and Computer Engineering *09/2019 - 07/2025*

University of Oklahoma, Norman, Oklahoma, USA

MSc, Telecommunication Engineering *08/2014 - 08/2016*

University of Engineering and Technology, Taxila, Pakistan

BSc, Telecommunication Engineering *11/2010 - 06/2014*

University of Engineering and Technology, Taxila, Pakistan

Relevant Graduate Coursework

Telecommunication Technology / LTE Networks | Emerging Topics in LTE-Advanced and 5G | Advanced Machine Learning / Computer Vision | Testbed Design for Emerging Cellular Networks | Numerical Analysis | Information Theory and Probability | Digital Signal Processing

Teaching, Research & Professional Experience

Teaching & Instruction

Lecturer, Department of Technology **12/2017 - 09/2019**

University of Lahore, Pakistan *Lahore, Pakistan*

- Taught undergraduate Electrical Engineering Technology courses including Computer Programming, Communication Systems, and Wireless Communication, connecting theory with implementation-oriented examples.
- Mentored and advised students on academic projects, technical problem solving, and career pathways, including senior-design and course-project development.
- Contributed to curriculum development and course-content alignment with program outcomes, industry needs, and accreditation requirements.
- Supported departmental accreditation documentation, examinations, student orientation, technical workshops, seminars, and academic administration.

Teaching areas: Wireless Communications | Digital and Analog Communication Systems | 5G/6G and Emerging Wireless Networks | Machine Learning and Artificial Intelligence | Signals and Systems | Digital Signal Processing | Computer Programming | Data Science and Experimental Methods

Student mentoring and research development

- Mentored undergraduate and graduate researchers on machine-learning implementation, wireless-network experimentation, simulation, data analysis, and technical writing.
- Supported Bachelor's and Master's thesis/project work in communications and wireless systems, including research methodology, experiment planning, manuscript preparation, and presentation.
- Advised students on senior/capstone projects, academic planning, and technical career development.
- Developed assignments, laboratory material, and implementation-oriented exercises connecting communication theory with practical engineering systems.

University Research Appointments

Graduate Research Assistant **09/2019 - 07/2025**

AI4Networks, University of Oklahoma *Tulsa, OK*

- Designed reusable Python simulation and data-generation platforms for large-scale experimentation, including a 3GPP-compliant network simulator and an AI-enabled 5G testbed.
- Built transfer-learning, adversarial-learning, GAN, autoencoder, and reinforcement-learning pipelines for sparse-data and cross-domain modeling.

- Developed robustness evaluation frameworks for models operating under distribution shifts, missing data, and limited labeled samples; led 100+ experiments from problem formulation through validation.
- Developed scalable self-healing and outage-management frameworks that improved modeled network resilience by 30%.
- Published peer-reviewed research, presented technical results internationally, and mentored junior researchers in ML implementation, experimental design, and technical writing.

Research Associate, ComSens Lab **06/2016 - 12/2017**
COMSATS University Islamabad, Pakistan *Islamabad, Pakistan*

- Conducted research in wireless communication systems and contributed to publication-oriented modeling, simulation, analysis, and technical writing.
- Mentored Bachelor's and Master's students on thesis projects, research methodology, experiment design, technical writing, and presentation of results.
- Reviewed research manuscripts and student technical reports, providing structured feedback aligned with publication and academic-quality standards.
- Assisted with research-proposal preparation, literature synthesis, lab seminars, paper-review meetings, and coordination of research activities.

Full-Time Research Scholar, Communication Systems Lab **08/2014 - 02/2016**
University of Engineering and Technology, Taxila, Pakistan *Taxila, Pakistan*

- Conducted full-time graduate research in communication systems and wireless engineering while completing the MSc in Telecommunication Engineering.
- Assisted faculty with Communication Systems and Digital Communication Systems instruction, student questions, assignments, grading, and exam preparation.
- Supervised laboratory sessions for Communication Systems and Wireless Communication and developed lab manuals and implementation-oriented exercises.
- Supported curriculum development and student technical-project activities connecting theoretical communication concepts with practical experimentation.

Industry Research & Engineering

Machine Learning Engineer **02/2025 - Present**
AT&T, contractor via Innovatix Inc. *Bedminster, NJ*

- Translate machine-learning research into operational telecommunications systems, emphasizing domain adaptation, reliability, model validation, and responsible decision support.
- Develop reusable cross-domain evaluation and error-analysis methods for AI systems operating under distribution shift and heterogeneous operational conditions.
- Collaborate across engineering, software, and public-safety teams to connect research methods with real-world network-assurance requirements.

Data Scientist and AI/ML Engineer **09/2024 - 12/2024**
Impac Exploration Services *Weatherford, OK*

- Built an LLM-enabled analytical pipeline for extracting and interpreting information from geochemical well data.
- Developed scalable ingestion, cleansing, transformation, and validation workflows using Python, Azure Databricks, Azure Functions, and Synapse Analytics.
- Automated ETL processing for large exploration datasets, reducing data-preparation time by 40% and improving reproducibility.
- Implemented cross-validation, data-quality, and consistency checks that improved reliability of analytical outputs by 20%.
- Collaborated with geologists to convert domain requirements and technical information into AI-ready datasets and actionable analytical outputs.

Advanced-Degrees Intern **06/2024 - 08/2024**
AT&T Labs *Austin, TX*

- Analyzed 500K+ LTE/5G-NR diagnostic logs and applied ML methods to detect anomalies, reducing diagnostic time by 20%.
- Investigated 5G NR-NTN cell-selection behavior and improved decision algorithms by 15% through data-driven analysis and validation.
- Conducted structured analysis of large operational datasets and communicated technical findings to engineering teams supporting next-generation wireless systems.

Advanced Wireless Technology Development Intern **09/2022 - 01/2023**
Futurewei Technologies *Schaumburg, IL*

- Improved autoencoder-based channel-state-information compression by 25% using transfer learning and systematic experimentation on large wireless datasets.
- Conducted 50+ experiments to validate model performance, robustness, and generalization under changing data conditions.
- Collaborated with research and engineering stakeholders to translate modeling results into recommendations for next-generation communication systems.

Advanced-Degrees Intern **06/2022 - 08/2022**
AT&T Labs *Bedminster, NJ*

- Designed ML and statistical methods for analyzing users' timing in 5G standalone networks, improving analytical precision by 20%.
- Analyzed large-scale user trace data with SQL and visualization tools and communicated findings to senior technical leadership.
- Converted raw operational traces into interpretable performance indicators to support diagnosis and network optimization.

Research Contributions

Publications

Real-World Medication Adherence Monitoring Using Hybrid Neural Networks and Smartwatches *2026*
HealthINF | Accepted

AI-Powered Resilience: A Dual-Approach for Outage Management in Dense Cellular Networks *2025*
Computer Communications

On Multi-Parameter Optimization and Proactive Reliability in 5G and Beyond Cellular Networks *2025*
Sensors 2025

An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin *2024*
IEEE PIMRC

Refining Wireless Propagation Models using Domain-Informed GANs amid Data Scarcity *2024*
IEEE VTC-Fall 2024

Data-Driven Intelligent Outage Management for High Shadowing Environments in 5G&B Networks *2024*
IEEE SmartNets

An AI-Enabled Framework to Defend Ingenious MDT-Based Attacks on the Emerging Zero Touch Cellular Networks *2024*
IEEE Network, 38(1), 228-237

Positioning Error Impact Compensation through Data-Driven Optimization in User-Centric Networks *2023*
IEEE GLOBECOM 2023

Toward a Hybrid RF/Optical Lunar Communication System (LunarComm) *2022*
IEEE Network

Towards Positioning Error Impact Characterization and Minimization in User-Centric RAN *2022*
IEEE WCNC

Unification of RF Energy Harvesting Schemes under Mixed Rayleigh-Rician Fading Channels *2020*
AEÜ - International Journal of Electronics and Communications

Research Projects & Platforms

SyntheticNET: 3GPP-Compliant Network Simulation *Research Project | 2020-2024*

Developed a Python-based, 3GPP-aligned simulation environment for realistic mobility, propagation, and cellular-network experimentation when measured data is sparse or costly. Supported controlled data generation and repeatable studies in propagation modeling, network optimization, digital twins, and resilient wireless AI.

Tools & Methods: Python, 3GPP, Wireless Simulation, Mobility Modeling, Propagation Modeling, Synthetic Data, Experiment Design, Digital Twins, Network Analytics

TurboRAN Experimental Platform *Research Infrastructure | 2020-2024*

Built and used a multi-band, multi-tier experimental platform for evaluating AI-enabled radio-access-network methods under controlled network conditions. Supported repeatable measurement collection, testbed experimentation, RAN optimization studies, and validation of emerging wireless-network algorithms.

Tools & Methods: 5G Testbeds, RAN, Network Measurement, Experiment Design, Wireless Prototyping, Multi-Band Systems, Performance Evaluation, RAN Optimization

Propagation Models & Digital Twins *Research Project | 2024*

Developed robust data-driven radio-propagation models for wireless digital twins using domain-informed generative learning and scarce measured data. Combined 3GPP-informed pretraining, GAN-based augmentation, domain adaptation, and statistical validation to improve resilience under distribution shift.

Tools & Methods: Propagation Modeling, Digital Twins, GANs, Domain Adaptation, 3GPP Models, Synthetic Data, Statistical Validation, Wireless AI, Distribution Shift

Wearable Intelligence for Health Monitoring *Research Project | 2024-2026*

Developed deep-learning workflows for recognizing medication-taking and dietary activities from smartwatch accelerometer and gyroscope signals in free-living settings. Evaluated windowing, feature extraction, CNN/LSTM-style sequence models, cross-user generalization, and recall-oriented performance for health-monitoring applications.

Tools & Methods: Wearable AI, Smartwatch IMU, Time Series, Deep Learning, CNN/LSTM, Feature Engineering, Cross-User Evaluation, Activity Recognition, Healthcare AI

AI for Telecom Network & Security Optimization *Certification Project | 2026*

Applied AI concepts to telecommunications network optimization, security analysis, and customer-experience improvement through a structured specialization project. Framed telecom use cases around operational data, AI-assisted analysis, optimization opportunities, and practical service-improvement recommendations.

Tools & Methods: AI for Telecommunications, Network Optimization, Telecom Security, Customer Experience, Python, TensorFlow, Operational Analytics, Service Improvement

Research Methods & Technical Expertise

Machine Learning: Deep learning, transfer learning, domain adaptation, adversarial learning, generative modeling, reinforcement learning, classification, time-series analysis, robustness evaluation, and model validation

Wireless & Networked Systems: LTE/4G, 5G-NR, NTN, mobility, RAN, propagation modeling, outage management, self-healing, network measurements, and service assurance

Research Methods: Experimental design, 3GPP-aligned simulation, statistical analysis, cross-validation, reproducible workflows, distribution-shift evaluation, technical writing, and publication development

Engineering Tools: Python, PyTorch, TensorFlow, scikit-learn, XGBoost, Pandas, NumPy, SQL, cloud analytics, data pipelines, visualization, Git, and Linux

Teaching & Mentoring: Wireless communications, communication systems, programming, machine learning, laboratory instruction, experimental methods, student projects, and research development

Academic Service & Recognition

Academic Service

- Contributed to curriculum development and course-content alignment with engineering program and industry needs.
- Supported departmental accreditation and quality-assurance documentation, examination processes, and academic administration.
- Reviewed research manuscripts and student technical reports and provided publication-oriented feedback.
- Participated in research seminars, paper-review meetings, student orientations, technical workshops, and academic events.
- Assisted research teams with proposal preparation, literature synthesis, experimental planning, and documentation.

Honors & Awards

William H. Barkow Scholarship *2024*
University of Oklahoma, Electrical and Computer Engineering

Clyde L. Farrar Endowment Scholarship *2024*
University of Oklahoma

P.H. Robinson Fellowship *2023*
University of Oklahoma

FAS-Charles Forgan Scholarship *2023*
University of Oklahoma

Nettie Vincent Boggs Graduate Endowment *2022*
University of Oklahoma

SHS-Paul D. Massad Scholarship *2022*
University of Oklahoma

Cleo Cross Memorial Scholarship *2021*
University of Oklahoma

Marjorie Haskell Engineering Scholarship *2021*
University of Oklahoma

Fully Funded PhD / Graduate Research Assistantship *2019*
University of Oklahoma

Self-Learning

1. Machine Learning Stanford Online / Coursera | Completed | Mar 29, 2020
Supervised and unsupervised learning, regularization, model evaluation, and practical predictive modeling.

2. Introduction to Deep Learning & Neural Networks with Keras IBM / Coursera | Completed | Aug 12, 2026
Neural networks, training workflows, optimization, loss functions, and model evaluation with Keras.

3. Introduction to Computer Vision with TensorFlow Google Cloud / Coursera | Completed | Aug 13, 2024
Image processing and convolutional neural networks for computer vision using TensorFlow.

4. AI Foundations & Industry Overview for Telecommunication Coursera | Completed | Jun 19, 2026
AI applications in telecom networks, operations analytics, service assurance, and automation.

5. Network and Security Optimization in Telecommunication Coursera | Completed | Jun 20, 2026
Data-driven network optimization, telecom security, reliability analysis, and operational efficiency.

6. 4G Network Fundamentals Institut Mines-Télécom / Coursera | Completed | Aug 2, 2026
LTE radio and core networks, cellular architecture, mobility management, and 4G fundamentals.

7. Generative AI Engineering and Fine-Tuning Transformers IBM / Coursera | Completed | Aug 6, 2026
Transformer architectures, model fine-tuning, embeddings, and generative-AI engineering.

8. 5G Network Fundamentals Institut Mines-Télécom / Coursera | In progress (21%)
5G NR radio, network architecture, service capabilities, and next-generation cellular fundamentals.